

PENETRATION TESTING REPORT

Footprinting, Reconnaissance & Network Scanning Phases

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program / Batch	B083D-Cybersecurity
Date	18/09/2026
Modules Completed	W2-PM1 (Multiple Kali Tools) W2-PM4 (theHarvester Footprinting) W2-PM5 (Zenmap Scanning)
Client / Target	1. Networkwalks (secured written permission already) 2. microsoft.com – theHarvester authorized training/lab exercise 3. My own local LAN network
Permission secured from client?	Yes
Phases Covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery

1. Liability Disclaimer

I have performed all of the activities described in this report only against systems, domains and devices for which I had secured written permission, or which I own myself. The theHarvester exercise against microsoft.com was carried out strictly as an authorized, non-intrusive OSINT training exercise using publicly available information sources, and did not involve any scanning, exploitation or unauthorized access of Microsoft infrastructure.

All material in this report is provided for education and research purposes only. Nothing in this document should be used to break the law. The instructor, the authors and Networkwalks are not responsible for how this knowledge is used – every action taken is the sole responsibility of the individual performing it. Misuse of these techniques can lead to criminal charges, heavy fines, loss of employment and a permanent criminal record. In most countries, unauthorized access to a computer system is a crime even when no damage is caused.

2. Introduction

This report covers three practical modules completed during Week 2 of my Cybersecurity & Ethical Hacking internship at Networkwalks:

- W2-PM1 – Footprinting the networkwalks.com domain using multiple Kali Linux tools
- W2-PM4 – OSINT gathering against microsoft.com using theHarvester (authorized training/lab exercise)
- W2-PM5 – Scanning my own local network with Zenmap

Together, these three modules show how a security professional moves from gathering public information about a target (footprinting and OSINT) to actively mapping the live hosts on a network (scanning).

The footprinting and theHarvester commands were run in Kali Linux, and the network scan was run on a Windows PC with Zenmap installed. Every step below includes the exact command or tool used, the result I observed, a screenshot placeholder for evidence, and a short note on why the finding matters from a security perspective.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers)
WhatWeb	Fingerprint web technologies (server, CMS, plugins, IP)

Tool	Purpose
nslookup	Resolve the domain name to its IP address using DNS
curl -I	Read the HTTP response headers of the website
wafw00f	Detect whether a Web Application Firewall protects the site
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV)
theHarvester	Collect publicly available emails, subdomains, hosts, IPs and ASN information about a target domain (OSINT)
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance (W2-PM1)

I performed reconnaissance against the *networkwalks.com* domain using six Kali Linux tools: **WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon**. Each tool was used to collect a different type of information about the target.

First, I used **WHOIS** to obtain publicly available domain registration information and identify the domain's name servers. The results provided information about the domain registration and hosting infrastructure.

I then used **WhatWeb** to identify technologies used by the website. The results identified the WordPress CMS and its active plugins, along with other information exposed by the website.

Using **Nslookup**, I resolved the domain name to its IP address.

I used **Curl** with the -I option to inspect the HTTP response headers. This provided additional information about the web application and exposed the WordPress REST API endpoint `/wp-json/`.

Next, I used **Wafw00f** to determine whether a Web Application Firewall was protecting the website.

Finally, I used **DNSRecon** to enumerate DNS records. The results provided information relating to name servers, mail servers, TXT records, service records and DNS software information.

4.2 theHarvester Footprinting (W2-PM4)

For this module, I used **theHarvester** from Kali Linux to gather publicly available OSINT information about *microsoft.com*, used strictly as an authorized training target for this exercise and separate from the *networkwalks.com* target used above.

I first ran theHarvester against the target using the Baidu data source. This single-source search did not return any results for this target:

Category	Finding
IP Addresses	0
Email Addresses	0
People	0
Hosts	0

I then re-ran theHarvester using all available public sources (-b all) to get a broader picture of what is publicly discoverable about the domain:

Category	Finding
ASNs	4
Interesting URLs	1
LinkedIn	No results
IP Addresses	138
Email Addresses	3
Hosts	9,968

The all-source scan returned a significantly larger volume of information than the single-source Baidu search. This shows how combining multiple OSINT sources gives a much more complete footprint of a target than relying on a single search engine or data source.

4.3 Network Scanning with Zenmap (W2-PM5)

For the third activity, I used Zenmap to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows ipconfig command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected Ping Scan to identify active hosts.

The example results provided in the practical identified four live hosts:

- 10.0.0.1
- 10.0.0.2
- 10.0.0.4
- 10.0.0.6

The example results also included four MAC addresses.

After completing the scan, I opened the Topology section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.

5. Risk Analysis / Impact

Based on the information collected during the footprinting, theHarvester and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified the CMS, version and active plugins in use	Attackers may use exposed technology/version information to identify software requiring further security review	● Medium
2	Server IP address identifiable	Nslookup resolved the domain to a public IP address	Provides information about the network location of the web service	● Low
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● Low
4	WAF technology identifiable	Wafw00f identified the WAF technology in use	Reveals information about the web application's security architecture	● Low
5	DNS infrastructure information exposed	DNSRecon identified name server, mail server and other DNS-related records	DNS information can help build a broader infrastructure profile	● Medium
6	OSINT data (emails, hosts, IP ranges) publicly harvestable	theHarvester identified email addresses, hosts, IP ranges and ASN information associated with the target domain from public sources	Harvested emails and hosts can be used to build phishing/social-engineering target lists or to expand the mapped attack surface	● Medium
7	Multiple live hosts visible on local network	Zenmap identified four live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	● Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting, OSINT and scanning exercises, not confirmed vulnerabilities.

These practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these three modules. Therefore, the presence of information such as a software version, IP address, harvested email address or

DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information** – Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
2. **Keep software updated** – CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers** – HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records regularly** – DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF** – Keep the WAF enabled and tuned, since it already blocks naive attacks.
6. **Review and limit publicly harvestable OSINT information** – Periodically run OSINT tools such as theHarvester against your own domain to see what emails, hosts and infrastructure details are publicly discoverable, and reduce or monitor what isn't necessary to expose.
7. **Perform regular internal network discovery** – Organizations should periodically scan their own networks to identify active devices.
8. **Investigate unknown devices** – Any unexpected device discovered during network scanning should be investigated and verified.
9. **Maintain network documentation** – Network topology and device information should be documented and updated regularly.
10. **Perform security testing with authorization** – Reconnaissance, OSINT gathering and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, OSINT-based reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

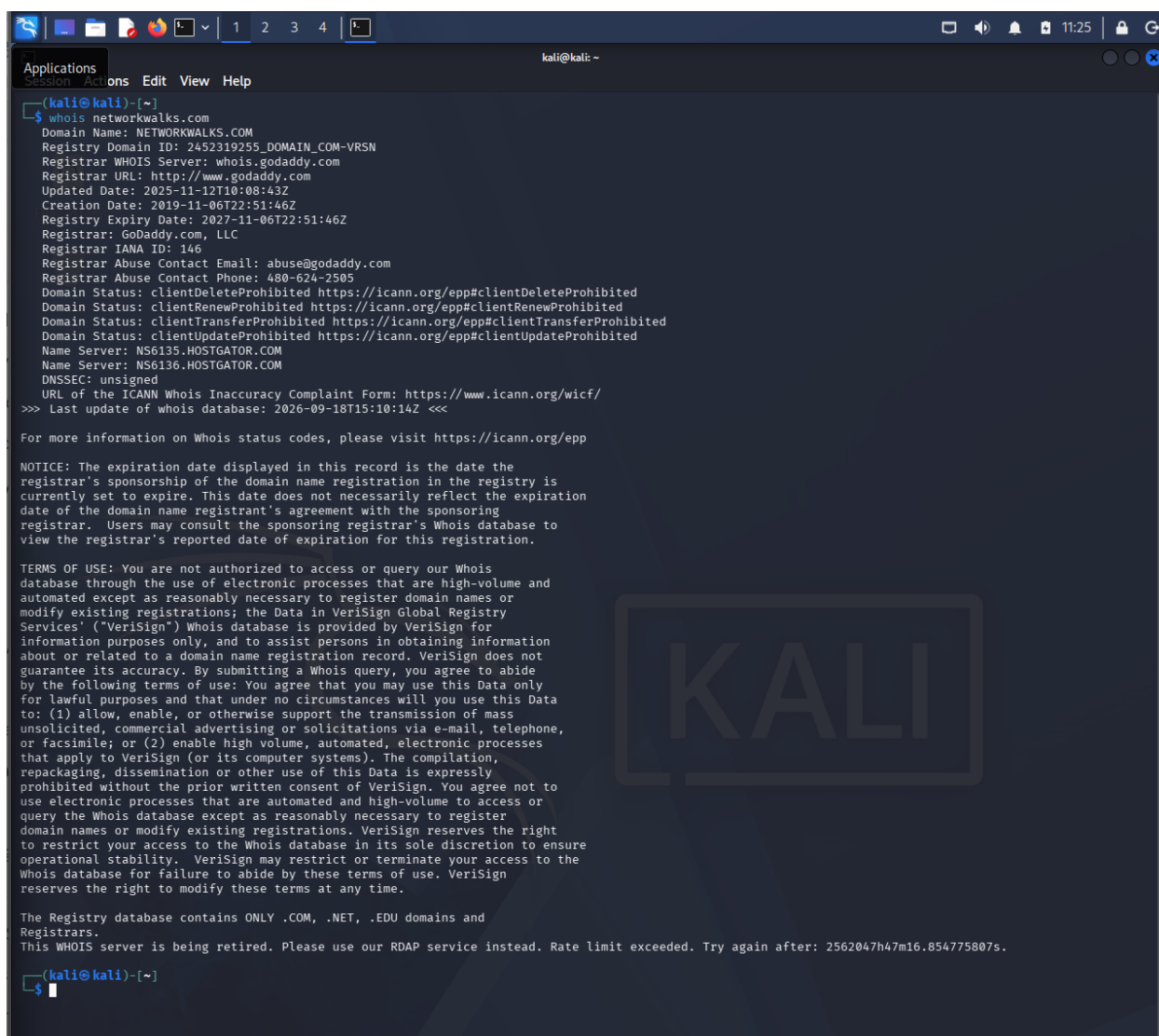
In the theHarvester activity, I learned how a single OSINT tool can pull together emails, hosts, IP ranges and ASN information from multiple public sources, and how the choice of data source – a single search engine versus an aggregated all-sources search – significantly changes the amount of information that is uncovered about a target.

In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology.

These exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses. I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance, OSINT gathering and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected



```

kali@kali: ~
$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-18T15:10:14Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

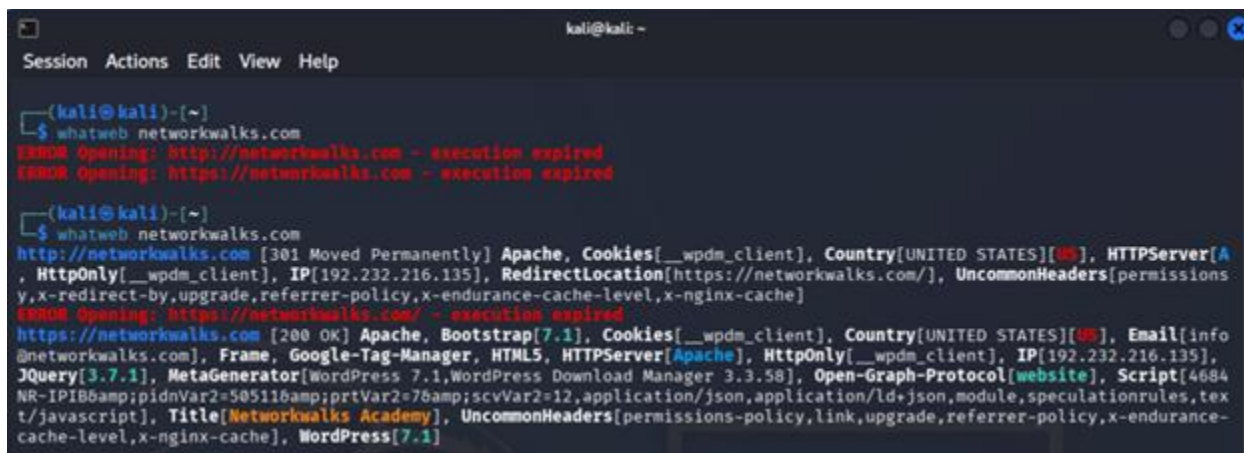
TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
operational stability. VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.

The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
This WHOIS server is being retired. Please use our RDAP service instead. Rate limit exceeded. Try again after: 2562047h47m16.854775807s.

kali@kali: ~
$

```

Screenshot 1 – WHOIS output



```

kali@kali: ~
$ whatweb networkwalks.com
ERROR Opening: http://networkwalks.com - execution expired
ERROR Opening: https://networkwalks.com - execution expired

kali@kali: ~
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][40], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
ERROR Opening: https://networkwalks.com/ - execution expired
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1], Cookies[__wpdm_client], Country[UNITED STATES][40], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPiB&pidnVar2=50511&prtVar2=76&pscVar2=12,application/json,application/ld+json,module,speculationrules,txt/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]

```

Screenshot 2 – WhatWeb output

```
(kali㉿kali)-[~]
$ nslookup networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

Screenshot 3 – Nslookup output

```
(kali㉿kali)-[~]
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=3f0c77c0daa309ff139b8c34e6fea66b; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Fri, 18 Sep 2026 16:30:07 GMT
server: Apache
```

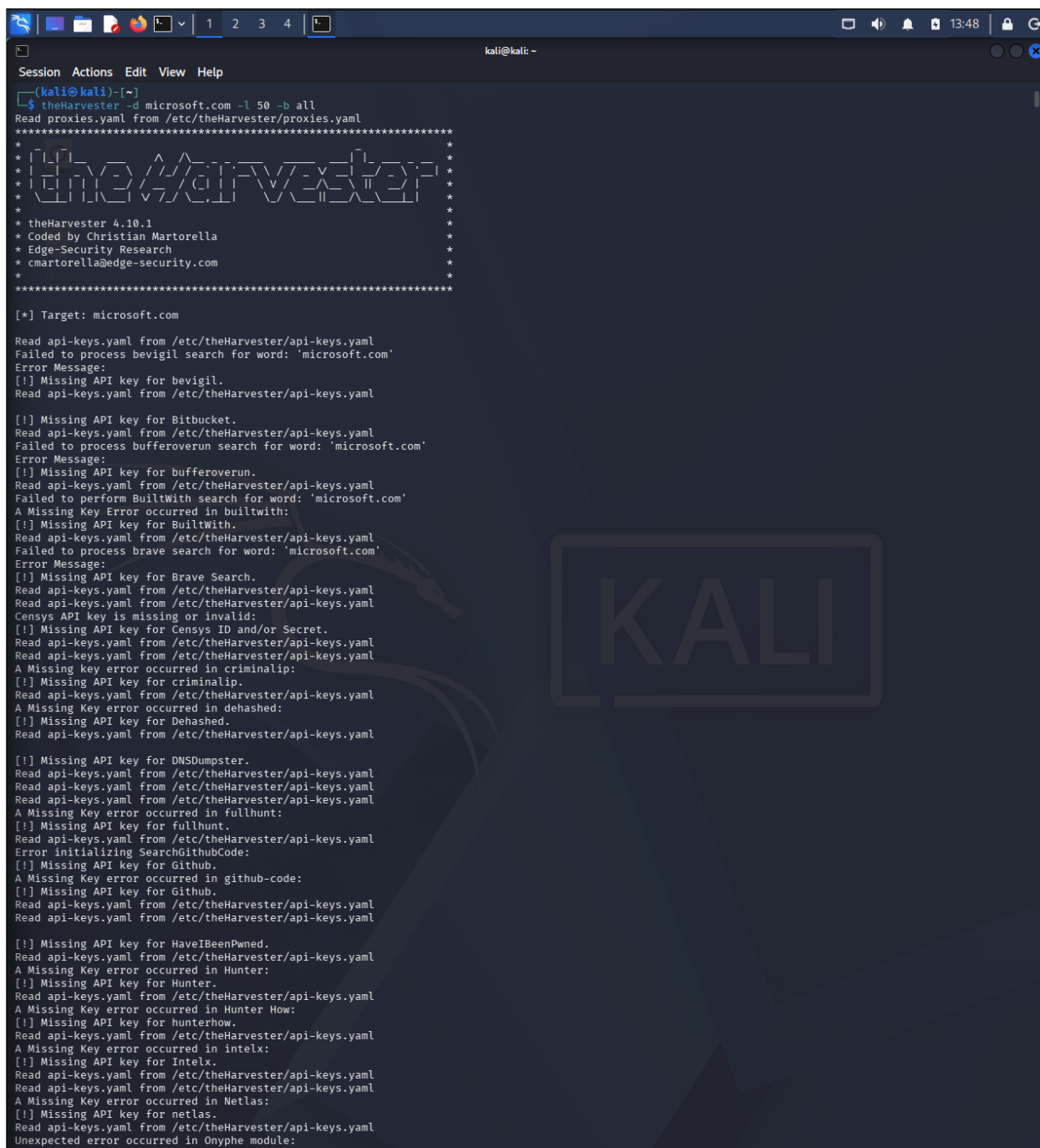
Screenshot 4 – Curl -I output

```
(kali㉿kali)-[~]
$ wafw00f networkwalks.com

      ( Woof! )
    /-----\
   /           \
  /             \
 /               \
/                 \
~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
```

Screenshot 5 – Wafw00f output



```
(kali@kali)~$ theHarvester -d microsoft.com -l 50 -b all
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****

[*] Target: microsoft.com

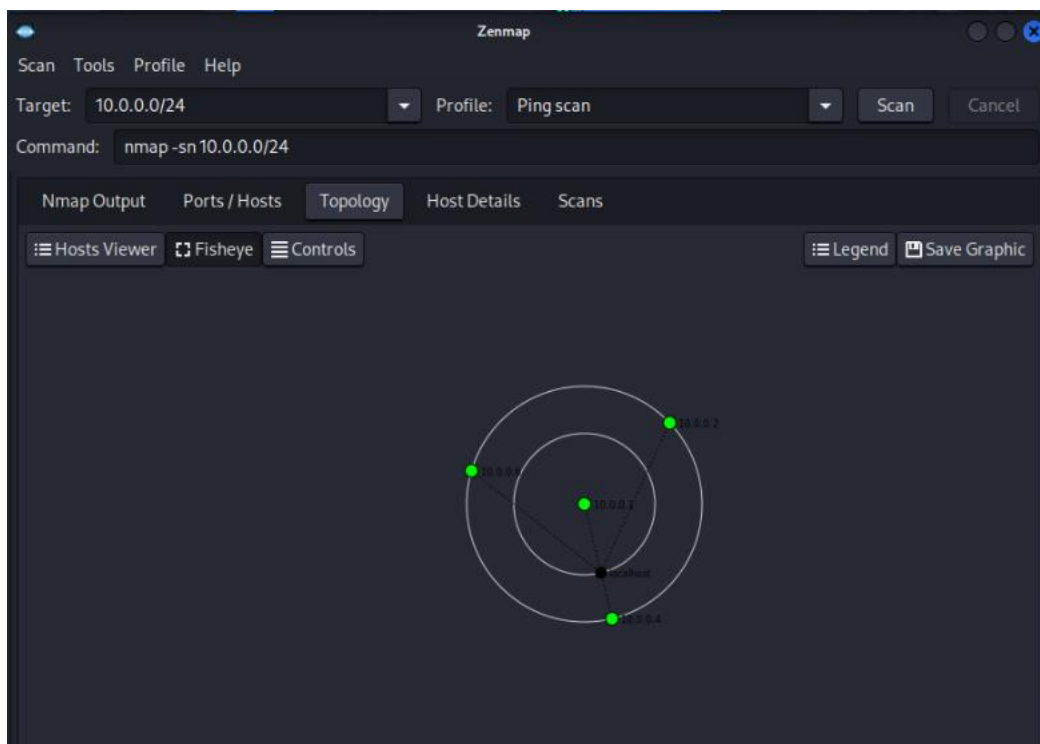
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverrun search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bufferoverrun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform BuiltWith search for word: 'microsoft.com'
A Missing Key Error occurred in builtwith:
[!] Missing API key for BuiltWith.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process brave search for word: 'microsoft.com'
Error Message:
[!] Missing API key for Brave Search.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Censys API key is missing or invalid:
[!] Missing API key for Censys ID and/or Secret.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing key error occurred in criminalip:
[!] Missing API key for criminalip.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in dehashed:
[!] Missing API key for Dehashed.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

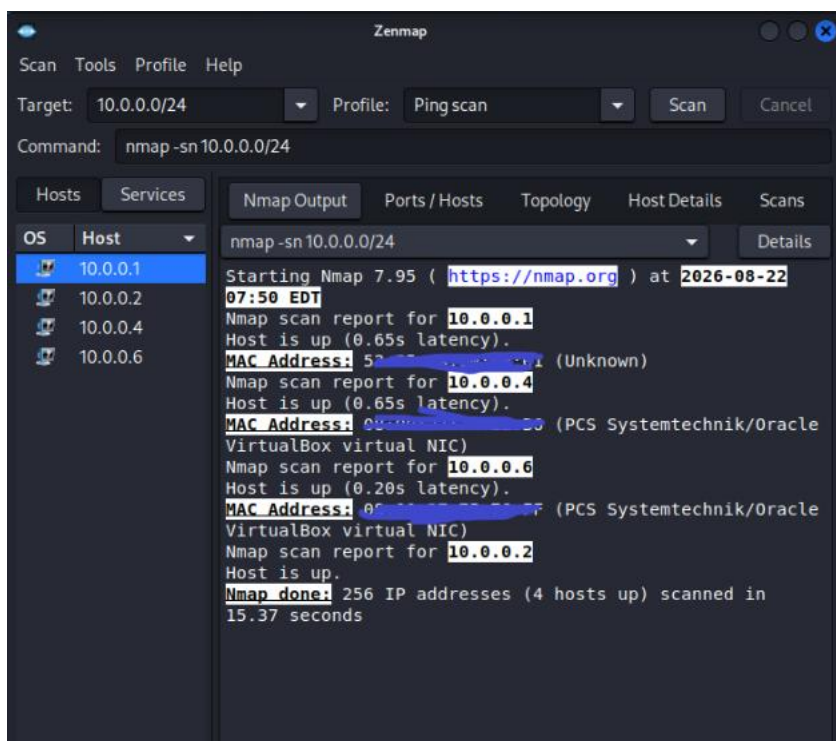
[!] Missing API key for DNSDumpster.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in fullhunt:
[!] Missing API key for fullhunt.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Error initializing SearchGithubCode:
[!] Missing API key for Github.
A Missing Key error occurred in github-code:
[!] Missing API key for Github.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for HaveIBeenPwned.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in Hunter:
[!] Missing API key for Hunter.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in Hunter How:
[!] Missing API key for hunterhow.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in intelx:
[!] Missing API key for Intelx.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in Netlas:
[!] Missing API key for netlas.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Unexpected error occurred in Onyphe module:
```

Screenshot 8 – theHarvester – All source results



Screenshot 9 – Zenmap – Network Topology



Screenshot 9 – Zenmap – Ping scan results

-End-

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